

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Winter Examination – 2022****Course: B. Tech.****Branch : E & T C****Semester : V****Subject Code & Name: BTETC501 - Electromagnetic Field Theory****Max Marks: 60****Date: 28/01/2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following:		12
A) Given point P (-2,6,3), and vector $\mathbf{A} = y \mathbf{a}_x + (x+z) \mathbf{a}_y$, Express P and A in Cylindrical and Spherical Coordinates.	Understand	6
B) Three field quantities are given as $\mathbf{P} = 2\mathbf{a}_x - \mathbf{a}_z$, $\mathbf{Q} = 2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z$, $\mathbf{R} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$. Determine (i) $(\mathbf{P} + \mathbf{Q}) + (\mathbf{P} - \mathbf{Q})$ (ii) $(\mathbf{Q} \cdot \mathbf{R} \times \mathbf{P})$ (iii) $(\mathbf{P} \times (\mathbf{Q} \times \mathbf{R}))$	Understand	6
C) Explain the term Curl of a vector and hence derive the Stokes Theorem	Understand	6
D) Explain the term Divergence of a vector field and hence derive the Divergence Theorem. Determine the Divergence of the vector field $\mathbf{A} = YZ \mathbf{a}_x + 4XY \mathbf{a}_y + Y \mathbf{a}_z$ at a given point (1,-2,3)	Understand	6
Q.2 Solve Any Two of the following.		12
A) Derive an Expression for Electric Field Intensity $\mathbf{E}$ due to Infinitely Long Straight Uniformly Charged Wire	Understand	6
B) State and explain : Ampere's Circuit law and Maxwell's Continuity equation	Remember	6
C) Find total charge enclosed by the volume with charge density $\rho_v = 4xyz^2$, For $0 \leq \rho \leq 2$, $0 \leq \phi \leq \pi/2$ and $0 \leq z \leq 3$.	Understand	6
D) State and explain the Coulomb's law and the superposition principle applied for the static charge distribution.	Remember	6
Q. 3 Solve Any Two of the following.		12
A) State and Explain the Boundary conditions for medium separating : Dielectric- Dielectric and Conductor – Dielectric	Remember	6
B) State and explain the gradient of a scalar field. Show that $\mathbf{E} = - (\text{Grad } V)$	Understand	6

where V is a scalar function and hence determine the gradient of the scalar field $U = x^2 y + x y z$

- C) State and explain the basic Maxwell's Equations in its point and integral form with their proper significance. **Remember** 6

Q.4 Solve Any Two of the following. 12

- A) A transmission line cable has following primary constants. $R = 75 \Omega/\text{km}$, $G = 60 \mu\text{mho}/\text{km}$, $L = 1.7 \text{ mH}/\text{km}$ and $C = 0.075 \mu\text{F}/\text{km}$, at a signal of 2kHz, calculate (i) characteristics impedance, (ii) attenuation constant (iii) phase constant (iv) wavelength (v) velocity and (vi) time taken by signal to travel 100 km along line. **Understand** 6
- B) Derive the expression for Electric Potential Energy and Electrical Potential. <https://www.batuonline.com> **Remember** 6
- C) Define the following terms with reference to EM Wave:
(i) Propagation constant (ii) characteristic impedance and (iii) VSWR **Remember** 6
- D) State ohms law in EM field and prove the equation for ohms law $J = \sigma E$, where J is conduction current density in amp per meter square, σ (Sigma) is the conductivity and E is electric field strength in volt per meter. **Understand** 6

Q. 5 Solve Any Two of the following. 12

- A) A uniform plane wave of 50 Hz propagating through Copper conductor having $\sigma = 6.8 \times 10^7 \text{ mho}/\text{m}$, $\epsilon_r = 1$, $\mu_r = 1$. Find (i) attenuation constant, (ii) phase constant, (iii) propagation constant, (iv) intrinsic impedance, (v) wavelength and (vi) phase velocity. **Understand** 6
- B) Given the circular plate parallel capacitor with radius of 5 cm where the distance of separation between the plates is 2cm and connected to a battery of 200 Volt. Find the value of capacitor and the energy stored in the capacitor. **Understand** 6
- C) Show that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within volume v minus the ohmic losses. **Understand** 6
- D) Derive the equation for the uniform EM plane waves for the free space. **Remember** 6

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